

16.3 Answers

① 6

② a) $f(x, y) = 3x^2 + 5xy + 2y^2 + C$

b) $\vec{F}$ is not conservative

c) $f(x, y) = ye^x + x \sin y + C$

③ $f(6, 5) - f(3, 2) = 150 - 12 = 138$

④ a) $f(x, y, z) = xyz + z^2$ (take $C=0$); 77

b) $f(x, y, z) = xy^2 \cos(z)$ (take $C=0$); 0